A11: Independent audit of Hopf spectral conditions and spectral simplicity

Directed Cyclic Networks research project

4 October 2026

1 Verdicts and exact-version disposition

GENERAL FREQUENCY / STRICT-CANDIDATE VERDICT	VERIFIED
SIMPLICITY / EIGENVECTOR / RESOLVENT VERDICT	VERIFIED
CANDIDATE-SURFACE VERDICT	VERIFIED
REALIZABILITY-CRITERION VERDICT	VERIFIED
FIVE-NODE CANDIDATE VERDICT	VERIFIED
SCOPE / P12-P13 HANDOFF VERDICT	VERIFIED
OVERALL VERDICT	VERIFIED

Required corrections: none. The six verdicts concern the precise spectral claims in P11 v1.0. They do not certify a parameter crossing, a Hopf bifurcation, a Lyapunov coefficient, a periodic orbit, a rotating wave, or a first loss of stability.

PROPOSED SCOPE CLARIFICATION RECOMMENDATION: ACCEPT AS PLANNING CLARIFICATION.

This recommendation is separate from the mathematical verdict. No frozen file is amended or silently superseded here.

Exact audited source. `proofs/P11_hopf_spectral_conditions.tex`, P11 v1.0, 3 October 2026, 25,914 bytes and 600 lines. Its full-file SHA-256 is

```
4be8e9b1e64c022b635f3cca9f7f49ffbc53
58a2250dda1715cb60e82d9b073b
```

The two displayed segments concatenate to the full hash. The manifest's last P11 CURRENT entry designates this exact file and hash. Earlier P11 placeholders in other snapshots are not audit inputs. All original files remain unchanged; the audit verdict does not promote the proof's lifecycle.

Method. This is an adversarial analytical reconstruction from the canonical field, current P02/P04/P05, and conditional P07 identities. Every source line is covered by the range ledger below. Exact and finite numerical checks are separate corroboration. The developer validation package was inventoried and hashed but its checker was neither read as a derivation nor executed. No literature search or later-module mathematics was undertaken.

2 Current-file and dependency authentication

The working root is `cyclic_network_project/`. Both copies of `cyclic_network_project_P11_draft.zip` have SHA-256

`8513c67effa28b825b52052e2c9333282f66`
`3b1031610ea150e58aeb16b58ad5.`

All 200 file members pass CRC and match the working files byte-for-byte. The P10 audited baseline has 199 members and SHA-256

`e0ecae17063fced8ed0a6ebb6fcf44456d5`
`c22065f51059274a0231659dcaaf5.`

Exactly five baseline members differ: P11, manifest, registry, compendium source and compendium PDF. The proposed clarification is the sole added member. All requested controls and prior proof/audit sources are present. The 31-item input inventory records names, full hashes, byte and line counts; proof mathematical-body hashes are also recorded. The 200-item baseline inventory permits a final preservation check independently of any lifecycle prose.

The authenticated controls are the manifest, claim registry, frozen theorem package, final novelty verdict, novelty decisions, open decisions, dependency map, proofs compendium, shared macros, canonical model, and notation/assumptions. Current proofs and audits are P01/A01, P02/A02, P03/A03, P04/A04, P05/A05, P07/A07, P08/A08, P09/A09 and P10/A10. Each current proof's computed full hash appears in its manifest closure. P01/P02 and P05/P08 retain their recorded E01 closures; P03/P04/P07/P09/P10 retain metadata-only closures and their body hashes. Original audits keep their historical verdicts. A02's source exists: its old missing-source header is superseded by the manifest's restoration record. A04/A05's historical validation-provenance qualifications are not erased.

Actual premises, not merely header declarations. The dependency map directly lists P02/P04/P05 $\to$ P11, and separately requires P07 for five-node use. The body respects this graph:

- P02 supplies the nonzero equilibrium equation, scalar polynomial, root classification and scalar bracket (lines 239–248, 475–490, 536–544).
- P04 supplies the oriented Fourier spectrum, conjugacy, normality, label-count multiplicity and uniform-mode identity (41–50, 141–166, 193–207, 245–260, 451–452, 537–539).
- P05 supplies complete selection, reachable representatives, strict cell endpoints and the concave equilibrium chart (54–76, 141–147, 291–300, 322–387).
- P07 is used only in the five-node subsection (391–398, 453–470). Its conditional role is preserved.

The inherited P03 coefficient formula is restated through P04, not used as a new direct DAG edge. P08/P09/P10 are authenticated context only. The two-positive-root argument actually factors P02's polynomial, so it does not conceal a P10 premise. P12/P13 are future recipients, and P14–P18 supply no results. The P13 placeholder is inspected solely as a convention requirement, not as a mathematical input. Elementary linear algebra and the local implicit/regular-level-set theorems supply the standard tools explicitly used below.

3 Layer A: general frequency and strict selection

The original predecessor convention is $(Pv)_i = v_{i-1}$ and $v_i^{(m)} = n^{-1/2} e^{2\pi \mathrm{i} mi/n}$. Thus the predecessor contributes $Ce^{-\mathrm{i}\theta_m}$, the positive successor-two term contributes $Ge^{2\mathrm{i}\theta_m}$, and the negative

predecessor-two term contributes $-He^{-2i\theta_m}$. The sign of the last imaginary contribution is positive. Consequently

$$r_m = d + C\cos\theta_m + (G-H)\cos 2\theta_m, \qquad \omega_m = -C\sin\theta_m + (G+H)\sin 2\theta_m.$$

These agree literally with current P04; no G/H phase reversal occurs.

P05 gives $1 \le j \le \lfloor (n+1)/4 \rfloor < n/2$ for every reachable nonuniform representative. The strict inequality follows from $n > 1$. For completeness, if $n = 4r$, the largest representative is $r < 2r$; if $n = 4r+2$, it is $r < 2r+1$. These cover every even size. If $n = 4r+1$ it is r, and if $n = 4r+3$ it is $r+1$, again strictly below $n/2$. Hence $0 < \theta_j < \pi$, $\sin\theta_j > 0$, and the two labels $j, n-j$ are distinct. The Nyquist singleton is never reachable.

Using $\sin 2\theta = 2\sin\theta\cos\theta$ gives independently

$$\omega_j = \sin\theta_j\{2(G+H)u_j - C\}, \qquad u_j = \cos\theta_j.$$

Since $\sin\theta_j > 0$, zero frequency is equivalent to $2(G+H)u_j = C$. For $u_j \le 0$ the bracket is at most $-C < 0$, so the frequency is strictly negative. For $u_j > 0$, the zero locus is exactly $G+H = C/(2u_j)$, with the claimed sign on either side. There is no decimal or asymptotic premise.

The definition requires $j \neq 0$, the complete selected set $\mathcal{S} = \{j\}$, $r_j = 0$ and $\omega_j \neq 0$. It excludes every maximal P05 tie, not every accidental equality among stable real parts. For every other representative t, completeness gives $r_t < 0$. Every Fourier label belongs to exactly one representative class, so all labels outside $L_j = \{j, n-j\}$ are strictly left. This includes label zero, all conjugate partners, and the even-n singleton. Repeated numerical eigenvalues among stable labels do not change the conclusion. P04 conjugacy then gives precisely $\lambda_j = i\omega_j$ and $\lambda_{n-j} = -i\omega_j$ on the imaginary axis.

GENERAL FREQUENCY / STRICT-CANDIDATE VERDICT: VERIFIED.

4 Layer B: simplicity, vectors, real plane and inverse gates

If a label outside L_j produced λ_j, its real part would be zero, contrary to strict selection. The other label in L_j gives the opposite value and is distinct because $\omega_j \neq 0$. Exactly one label produces λ_j. Interchanging the two labels gives the same result for λ_{n-j}. P04's multiplicity theorem makes each value algebraically and geometrically simple over $\mathbb{C}$. There are two simple eigenvalues, not one eigenvalue of multiplicity two.

There are $n-2 \ge 3$ other labels. This finite nonempty set has a strictly negative maximum real part, so

$$\eta = -\max_{\ell \notin L_j} r_\ell > 0, \qquad \Re\lambda_\ell \le -\eta \quad (\ell \notin L_j).$$

The critical Fourier span is conjugation-invariant and has a real form of dimension two. The complementary Fourier span is likewise conjugation-invariant, with real dimension $n-2$, and is stable. It may contain repeated stable eigenvalues; diagonalizability remains valid.

Both 0 and $2i\omega_j$ have zero real part, excluding equality with every stable eigenvalue. Zero differs from both critical values. Equality $2i\omega_j = i\omega_j$ would require $\omega_j = 0$; equality $2i\omega_j = -i\omega_j$ would require $3\omega_j = 0$. Thus both J^{-1} and $(2i\omega_j I - J)^{-1}$ exist, for either sign of ω_j. No nonlinear formula enters this argument.

Adjoint and basis audit. Write $J = Q\Lambda Q^*$ with unitary Q. Then $J^* = Q\overline{\Lambda}Q^*$; for $v_H = v^{(j)}$ one has

$$Jv_H = i\omega_j v_H, \qquad J^* v_H = -i\omega_j v_H, \qquad v_H^* v_H = 1.$$

Thus $p_H = v_H$ is correct for the actual pairing $\langle p, v\rangle = p^* v$. This also agrees with the current B07 resolution of the opposite-sign adjoint equation. The TODO P13 source requires reconciliation of this pairing and Taylor factors but states no competing adopted formula. There is no convention mismatch; P13 still owns its Taylor, factorial, time and coefficient normalizations.

Put $v_H = A + \mathrm{i}B$. Orthogonality of v_H and $\overline{v}_H$ yields $v_H^{\mathsf{T}} v_H = 0$, while $v_H^* v_H = 1$. Taking real and imaginary parts gives

$$\|A\|^2 - \|B\|^2 = 0, \quad A^{\mathsf{T}} B = 0, \quad \|A\|^2 + \|B\|^2 = 1.$$

Therefore $e_c = \sqrt{2}A, e_s = \sqrt{2}B$ are orthonormal. Real and imaginary parts of the eigenvector equation give

$$Je_c = -\omega_j e_s, \quad Je_s = \omega_j e_c, \qquad [J]_{(e_c, e_s)} = \begin{pmatrix} 0 & \omega_j \\ -\omega_j & 0 \end{pmatrix}.$$

The columns have exactly the stated orientation. For a nonzero coefficient vector (z_c, z_s) its angle has derivative $(z_c \dot{z}_s - z_s \dot{z}_c)/(z_c^2 + z_s^2) = -\omega_j$. This is a linear coefficient-plane statement, not nonlinear propagation.

Excluded degeneracies. Two different representative classes at real part zero supply additional critical labels, whether or not their numerical eigenvalues coincide. The single-simple-pair hypothesis fails. P11 gives no named multiple-criticality bifurcation classification. If a genuine pair has $r_j = \omega_j = 0$, both labels produce zero. P04 makes this eigenvalue semisimple, with algebraic and geometric multiplicity equal to the total number of zero labels. Under strict representative selection that number is exactly two and the real kernel dimension is two. Without strictness, all other zero labels must be counted. Neither case is admitted as a strict spectral Hopf candidate.
SIMPLICITY / EIGENVECTOR / RESOLVENT VERDICT: VERIFIED.

5 Layer C: regular candidate surfaces

P02's nonzero equilibrium relation gives precisely P05's open chart

$$a = (k + h)x - \frac{c - g}{1 + x} > 0, \qquad x, k, c, g, h > 0.$$

Substitution into the unsimplified diagonal coefficient gives

$$\begin{aligned} d &= a - 2kx - \frac{cx}{(1+x)^2} - \frac{g}{1+x} - hx \\ &= -kx - \frac{c}{1+x} - \frac{cx}{(1+x)^2} = -kx - \frac{c(1+2x)}{(1+x)^2}. \end{aligned}$$

Let $v_j = 2u_j^2 - 1 = \cos 2\theta_j$, a scalar distinct from $v^{(j)}$. It follows that

$$R_j = -kx - \frac{c(1+2x)}{(1+x)^2} + \frac{cu_j}{1+x} + x\left(\frac{g}{(1+x)^2} - h\right) v_j = r_j.$$

At fixed x, c, g, h, $\partial_k R_j = -x < 0$, so $R_j = 0$ is the regular graph

$$k = k_H = \frac{1}{x}\left[-\frac{c(1+2x)}{(1+x)^2} + \frac{cu_j}{1+x} + x\left(\frac{g}{(1+x)^2} - h\right) v_j\right].$$

Physical admissibility still requires $k_H > 0$ and $a_H = (k_H + h)x - (c - g)/(1 + x) > 0$. These restrictions cannot be dropped. Inside a strict selected cell, nonzero frequency is an additional open restriction on this regular level set. The claim is not that every reachable cell contains a nonempty portion; Layer D addresses exactly that.

P05 proves that $G - H \geq 0$ permits only the uniform maximizer. Therefore any nonuniform selected representative forces $D_c = H - G > 0$ and $h - g/(1 + x)^2 = D_c/x > 0$. For the map $\Phi(x, k, c, g, h) = (a, k, c, g, h)$ the determinant is

$$\det D\Phi = k + h + \frac{c - g}{(1 + x)^2} = k + \frac{D_c}{x} + \frac{c}{(1 + x)^2} > 0.$$

The local inverse function theorem transfers the regular surface to physical parameter space along the specified branch. This does not identify different branches globally and does not prove global injectivity. The derivative in k above lets a follow the chart; it is not a crossing derivative along the later fixed-a path.

For $u_j > 0$,

$$\partial_k \omega_j = 0, \qquad \partial_g \omega_j = \frac{2x u_j \sin\theta_j}{(1 + x)^2} > 0.$$

The determinant of the (k, g) minor of $D(R_j, \omega_j)$ is

$$-\frac{2x^2 u_j \sin\theta_j}{(1 + x)^2} \neq 0.$$

Any physically admissible nonempty joint level set is locally codimension two in branch coordinates and codimension one in $R_j = 0$. Physical transfer is conditional on the same concave local chart. For $u_j \leq 0$ the joint set is empty by the strict sign already proved. A statement of rank does not itself assert nonemptiness or a bifurcation unfolding.

CANDIDATE-SURFACE VERDICT: VERIFIED.

6 Layer D: independent reconstruction of realizability

Fix a reachable nonuniform representative and write

$$w_j = 1 - 2u_j^2, \quad L = \max(0, \beta_j), \quad U = \beta_{j-1}, \quad B = \frac{w_j}{4(1 - u_j)}, \quad z = \frac{C}{4D_c}.$$

P05's cosine grid is strictly decreasing. Its positive strict cell is $L < z < U$. The endpoint β_j exists because $j \leq j_{\max} < N = \lfloor n/2 \rfloor$. Reachability gives $U > 0$; strict midpoint decrease gives $L < U$. In particular a zero lower midpoint is excluded by $z > 0$, not included as an attainable tie.

Necessity from positive physical coefficients. Use $c = C(1 + x)$ and $G - H = -D_c$ in the independently derived R_j. Then $R_j = 0$ is equivalent to

$$kx = D_c w_j - C\left(1 - u_j + \frac{x}{1 + x}\right), \qquad \frac{kx}{C} = \frac{w_j}{4z} - 1 + u_j - \frac{x}{1 + x}.$$

Since $k, x, C > 0$ and $u_j < 1$, this forces

$$w_j > 0, \qquad \frac{w_j}{4z} > 1 - u_j + \frac{x}{1 + x}.$$

The first condition is exactly $|u_j| < 1/\sqrt{2}$. Equality $w_j = 0$ cannot work with $k > 0$. Dropping the positive term $x/(1 + x)$ yields $z < B$. Combining this with $z > L$ gives $L < B$. Thus every physical strict spectral candidate must satisfy both claimed inequalities.

Sufficiency, including all physical restrictions. Assume $w_j > 0$ and $L < B$. Both U and B exceed L, so choose

$$L < z < \min(U, B).$$

Then $\delta = w_j/(4z) - (1 - u_j) > 0$. For example, $x = \delta/[2(1 + \delta)] > 0$ satisfies $x/(1 + x) < \delta$; this makes the fixed-vertex inequality strict. More generally, for any prescribed x, z satisfying it, choose any $C > 0$, put $D_c = C/(4z)$, and set

$$k = \frac{D_c w_j - C(1 - u_j + x/(1 + x))}{x} > 0.$$

For a free $G > 0$, define

$$c = C(1 + x), \qquad g = \frac{G(1 + x)^2}{x}, \qquad h = \frac{G + D_c}{x},$$
$$a = kx + D_c - C + G\left(1 + \frac{1 + x}{x}\right).$$

The first four coefficients are positive. Their effective coefficients are exactly $C, G, H = G + D_c$, hence the selected vertex is still z. Directly,

$$(k + h)x - \frac{c - g}{1 + x} = kx + G + D_c - C + \frac{G(1 + x)}{x} = a.$$

This proves the equilibrium equation, and the definition of k proves $R_j = 0$. The coefficient of G in a is positive, so sufficiently large G gives $a > 0$ without moving k, z, x or R_j.

The frequency bracket is

$$2(G + H)u_j - C = 2(D_c + 2G)u_j - C.$$

For $u_j \le 0$ it is strictly negative. For $u_j > 0$ its only possible zero is

$$G_0 = \frac{C}{4u_j} - \frac{D_c}{2}.$$

At most one positive value is forbidden; arbitrarily large positive G with $a > 0$ can avoid it. This proves sufficiency both for the fixed-x, z statement and for the full-cell statement. All parameters remain finite; no limit point at $x = 0$ is used as a physical realization. The necessary-and-sufficient criterion is therefore exactly

$$\boxed{w_j > 0 \quad \text{and} \quad \max(0, \beta_j) < \frac{w_j}{4(1 - u_j)}.}$$

Exact obstruction checks. For $n = 8, j = 1$, $u_1 = \cos(\pi/4) = 1/\sqrt{2}$ and $w_1 = 0$. The equality for kx would have a negative right side. The cell can realize dominance, but no physical zero spectral abscissa in that cell. For $n = 12, j = 2$, $u_2 = 1/2$, $u_3 = 0$, and

$$w_2 = \frac{1}{2}, \qquad \beta_2 = \frac{u_2 + u_3}{2} = \frac{1}{4}, \qquad L = \frac{1}{4} = B.$$

The upper endpoint is $\beta_1 = (\sqrt{3} + 1)/4 > 1/4$; the cell is nonempty. Every strictly interior vertex has $z > 1/4$, contrary to the necessary $z < B = 1/4$. Even the lower boundary would require $x/(1 + x) < 0$ for positive k. These are genuine obstructions, not a midpoint indexing error. Neither contradicts P05's dominance realization. P11 replaces, rather than retains, the false optional universal assertion.

REALIZABILITY-CRITERION VERDICT: VERIFIED.

7 Layer E: exact five-node candidate and spectral handoff

Set $\phi = (1+\sqrt{5})/2$ and $\sigma = \sqrt{10-2\sqrt{5}}/4 = \sin(\pi/5) > 0$. Then $\phi^2 = \phi + 1$, $\phi^{-1} = \phi - 1$, $1 < \phi < 2$. For

$$x_H = 1, \quad c = 2, \quad g_H = 16, \quad h = 6, \quad k_H = \frac{3\sqrt{5}-5}{4} = \frac{3\phi-4}{2}, \quad a_H = \frac{47+3\sqrt{5}}{4} = \frac{3\phi+22}{2},$$

positivity of k_H follows from $5/3 < \sqrt{5}$, because $25/9 < 5$. Positivity of the other four coefficients is immediate. In the original canonical equation, before polynomial reduction,

$$F_i(\mathbf{1}) = a_H - k_H + \frac{2}{2} - \frac{16}{2} - 6 = a_H - k_H - 13 = 0.$$

This verifies all five components of the actual DM field.

The effective values are $s = 2, C = 1, G = 4, H = 6, T = -2, S = 10$, with

$$d = a_H - 2k_H - \frac{1}{2} - 8 - 6 = -k_H - \frac{3}{2} = \frac{1-3\phi}{2}.$$

P07's exact phase data are $u_1 = (\phi-1)/2$, $u_2 = -\phi/2$, $\sin\theta_1 = \phi\sigma$, $\sin\theta_2 = \sigma$, $\sin 2\theta_1 = \sigma$, $\sin 2\theta_2 = -\phi\sigma$. Substitution, without using only an interval table, gives

$$\begin{aligned} r_1 &= \frac{1-3\phi}{2} + \frac{\phi-1}{2} + \phi = 0, \\ r_0 &= \frac{1-3\phi}{2} + 1 - 2 = -\frac{3\phi+1}{2}, \\ r_2 &= \frac{1-3\phi}{2} - \frac{\phi}{2} - (\phi-1) = -\frac{3\sqrt{5}}{2}. \end{aligned}$$

Their difference $r_0 - r_2 = (3\phi-4)/2 = k_H > 0$ proves $r_1 > r_0 > r_2$ exactly. The imaginary parts are

$$\omega_H = -\phi\sigma + 10\sigma = \sigma(10-\phi) > 0, \qquad \omega_2 = -\sigma - 10\phi\sigma.$$

The exact inequality $10 - \phi > 8$ certifies nonzero frequency. Thus

$$\begin{aligned} \lambda_0 &= -\eta_H, & \eta_H &= \frac{3\phi+1}{2}, \\ \lambda_1 &= \mathrm{i}\omega_H, & \lambda_4 &= -\mathrm{i}\omega_H, \\ \lambda_2 &= -\frac{3\sqrt{5}}{2} - \mathrm{i}\sigma(1+10\phi), & \lambda_3 &= \overline{\lambda_2}. \end{aligned}$$

The general strict criterion proves individual simplicity of labels $1, 4$ and excludes all other imaginary-axis labels. The gap is exactly $-\max(r_0, r_2) = \eta_H$, not merely a lower estimate.

Upper-root identity from P02 only. Put $A = k_H + h = (3\phi+8)/2$, $b = a_H - A = 7$ and $\mu = a_H + 2 - 16 = (3\phi-6)/2 < 0$. Since $A - b = \mu$,

$$Q(X) = AX^2 - bX - \mu = A(X-1)(X-x_-), \qquad x_- = -\frac{\mu}{A} = \frac{3(2-\phi)}{3\phi+8}.$$

The numerator and denominator are positive, and their difference is $6\phi+2 > 0$, proving $0 < x_- < 1$. The discriminant equals $A^2(1-x_-)^2 > 0$. Thus there are two distinct positive roots and 1 is precisely the upper root. Merely knowing $b > 0, \mu < 0$ would not have established this without checking the discriminant; P11 checks it.

Local branch, without a path derivative. Freeze $\pi(g) = (a_H, k_H, 2, g, 6)$, with $g_H = 16$ and $j = 1$. The scalar bracket is

$$B(X, g) = a_H - (k_H + 6)X + \frac{2 - g}{1 + X}.$$

At $(1, 16)$, $B = 0$ and

$$B_X = -A - \frac{2 - 16}{4} = -A + \frac{7}{2} = -\frac{3\phi + 1}{2} = -\eta_H < 0.$$

P04's uniform eigenvalue is $f_X = B + XB_X$; the equality $B_X = \lambda_0$ here uses both $B = 0$ and the particular state $X = 1$. It is not an incorrectly generalized identity for arbitrary X. The scalar implicit-function theorem gives a unique smooth local root near 1, with positive state and parameters. Distinctness, positivity and ordering of the two roots persist by continuity, so this is $x_+(g)$. No dx_+/dg or eigenvalue derivative along this path is computed.

Exact vectors and resolvents. The required vector is the positive-exponent Fourier column

$$v_H = \frac{1}{\sqrt{5}}(1, e^{2\pi \mathrm{i}/5}, e^{4\pi \mathrm{i}/5}, e^{6\pi \mathrm{i}/5}, e^{8\pi \mathrm{i}/5})^{\mathsf{T}}, \qquad p_H = v_H.$$

With node order $0, \ldots, 4$, Layer B verifies $J_H v_H = \mathrm{i}\omega_H v_H$, $J_H^* p_H = -\mathrm{i}\omega_H p_H$ and $p_H^* v_H = 1$. Its real basis has precisely the stated block and coefficient-plane angular velocity $-\omega_H$. The complete orthogonal spectral projectors give

$$J_H^{-1} = \sum_{m=0}^{4} \lambda_m^{-1} v^{(m)} v^{(m)*},$$

$$(2\mathrm{i}\omega_H I - J_H)^{-1} = \sum_{m=0}^{4} (2\mathrm{i}\omega_H - \lambda_m)^{-1} v^{(m)} v^{(m)*}.$$

All denominators are nonzero by Layer B. In particular,

$$\lambda_0 = -\eta_H \neq 0, \qquad 2\mathrm{i}\omega_H - \lambda_2 = \frac{3\sqrt{5}}{2} + \mathrm{i}\sigma(21 + 8\phi) \neq 0,$$

because $2(10 - \phi) + (1 + 10\phi) = 21 + 8\phi$. This supplies spectral data only; no nonlinear interaction channel is evaluated.
FIVE-NODE CANDIDATE VERDICT: VERIFIED.

8 Layer F: literal ledger, scope and planning recommendation

Item	Independent disposition
H1	Established: all five physical coefficients are strictly positive.
H2	Established: each original-field component vanishes exactly at **1**.
H3	Established: the fixed g-path has a unique local positive upper root near 1.
H4	Established: labels $1, 4$ give the positive-frequency pair $\pm\mathrm{i}\omega_H$.
H5	Established: both numerical critical eigenvalues are separately simple.

Item	Independent disposition
H6	Established: every other eigenvalue has real part at most $-\eta_H < 0$.
H7	Established: zero is outside the spectrum.
H8	Established: $2\mathrm{i}\omega_H$ is outside the spectrum.
H9	Established: exact normalized right/adjoint vectors and real-plane orientation.
H10	Correctly unestablished: transversality; no state or eigenvalue derivative along g.
H11	Correctly unestablished: first Lyapunov coefficient and nonlinear nondegeneracy.
H12	Correctly unestablished: periodic orbit, branch stability and rotating wave.

The branch-coordinate k derivative and the partial frequency derivative in the surface proof do not assert H10. The scalar B_X derivative proves regularity of an equilibrium branch; it is not dx_+/dg. No second or third field derivative, normal-form tensor, Fourier nonlinear interaction, Lyapunov value/sign, or periodic-branch stability appears in P11. The word Hopf is consistently qualified as spectral candidate or future package. No earlier paragraph overrides H10–H12.

Separate clarification review. The proposed note correctly identifies frozen attempt-order row 10, "certify first crossing and exclusions," and novelty-decisions S11, which seeks an open first-loss regime along a specified path. The latter also requires a nonempty regime, a regular positive branch, complete spectral exclusions, and later transversality/nonlinear nondegeneracy. An isolated exact point and a regular spectral surface do not by themselves complete that target.

The proposed clarification allocates P11 the spectral candidate, its regular branch and exclusion data; P12 the moving-equilibrium crossing derivative; and P13 the nonlinear nondegeneracy and assembled Hopf theorem. A first-loss assertion still requires an explicit interval/path statement. This is consistent with the actual dependency map, which already assigns spectral conditions to P11, transversality to P12, and the assembled theorem to P13. The exact five-node point and Layer C prove the asserted nonempty regular candidate surface locally. The cell criterion refines an optional existence question and does not retract any P05 theorem.

Adoption would clarify the allocation and wording of planning obligations; it would not change direct or conditional dependency edges, canonical conventions, novelty ratings, the assembled first-Hopf target, or currently permitted manuscript claims. The stronger first-loss/open-regime target remains unproved and is not discharged by this audit. Exact-model priority remains unclosed under the existing bounded novelty record. No new novelty assessment or literature survey is made. The clarification should remain an explicit planning action at closure, rather than being treated as a mathematical theorem or an automatic frozen-file edit.

Recommendation: ACCEPT AS PLANNING CLARIFICATION.
SCOPE / P12-P13 HANDOFF VERDICT: VERIFIED.

9 Adversarial checks and validation provenance

The attempted failure modes were treated analytically, not merely by searching for numerical counterexamples:

1. Pair reachability: all four residue classes of n and the Nyquist endpoint were checked in Layer A.

2. Extra imaginary labels: complete representative selection covers all labels, including uniform and even singleton labels.

3. False simplicity: label equality was separated from numerical-value equality; the latter is excluded using strict real parts and nonzero frequency.

4. A $2i\omega$ pole: stable and both critical values were checked separately.

5. Wrong adjoint sign: the actual unitary diagonalization and Hermitian pairing give the opposite-sign adjoint equation.

6. Failed physical projection: strict nonuniform selection forces $D_c > 0$, making the chart determinant strictly positive.

7. Missing realizability restrictions: necessity includes $k > 0$, and sufficiency separately enforces $a > 0$ and avoids the sole possible frequency zero.

8. The $(8, 1)$ obstruction survives the boundary $w = 0$ exactly.

9. The $(12, 2)$ obstruction uses β_2, not β_1; equality is insufficient.

10. The five-node tuple is substituted into the original field first.

11. Every five-node eigenvalue and imaginary sign is reconstructed from exact phase data, then independently checked in the direct matrix.

12. The upper-root identity is established by exact factorization and positive discriminant, without P10.

13. The proposed note is checked against the actual stronger planning language; accepting it does not certify first loss or reduce the assembled target.

An explicit physical frequency-degenerate check. This independent exclusion test uses $n = 5, j = 1$, $x = 1/10$, $C = 1$, $D_c = 1$, $G = (\phi - 1)/2$ and $z = 1/4$. The strict cell lower endpoint is zero and its upper endpoint is $(1 + \phi)/4$, so this vertex is strict. Here $u = (\phi - 1)/2$, $w = \phi/2$ and

$$kx = \phi - \frac{35}{22} > 0,$$

since $\sqrt{5} > 24/11$ follows by squaring $(605 > 576)$. Use the physical construction in Layer D. Then $a = kx + 12G > 0$, all other coefficients are positive, $R_1 = 0$, but $2(D_c + 2G)u - C = 2\phi u - 1 = 0$. There is an exact two-dimensional real kernel with stable complement, not a strict spectral Hopf candidate. This demonstrates a genuine physical excluded point, not a fabricated free-coefficient example. It also witnesses that the conditional rank-two statement need not be vacuous.

An explicit critical tie check. For $n = 20$, the neighboring representatives $4, 5$ have cosines $u_4 = (\phi - 1)/2$ and $u_5 = 0$. Choose $C = 1$, $D_c = \phi$, $z = 1/(4\phi) = (u_4 + u_5)/2$, $x = 1/10$ and $G = 1$. Then $kx = \phi - 12/11 > 0$ and the same physical construction gives $a = 2\phi + 109/11 > 0$. Both representative real parts equal zero, and P05's midpoint rule excludes other representatives. Four Fourier labels are critical. This fails strict selection regardless of numerical frequency coincidences; no named bifurcation is inferred.

Independent exact arithmetic. The new checker uses rational arithmetic in $\mathbb{Q}(\phi, t)$ with $\phi^2 = \phi + 1$ and $t^2 = (\phi - 3)/4$, interpreted as $t = \mathrm{i}\sigma$. The fifth root is $z = (\phi - 1)/2 + \phi t$. Checking unnormalized columns $w_i^{(m)} = z^{im}$ gives $w^{(m)*} w^{(m)} = 5$; division by $\sqrt{5}$ yields the project normalization exactly. The script performs 144 exact assertions, including the original residual, direct 5×5 matrix eigenvectors, Gram matrix, adjoint residuals, real basis identities, both spectral-inverse products, mode denominators, branch polynomial and discriminant identities, exact obstruction constants, and two exact null columns at the excluded point. Exact inequalities are proved above; floating-point readings are not used to certify them.

Finite corroboration and limits. With Python 3.12.14 and NumPy 2.3.5, the separate checker examined all 646 reachable nonuniform cells for $5 \le n \le 72$: 179 satisfy the criterion and 467 do not. It constructed 358 strict physical candidates, ran 9,340 obstruction probes, and constructed 111 physical critical ties. A separately seeded raw-parameter search made 1,500 trials and found 147 admissible strict candidates, all consistent with necessity. There were 206 direct matrix-spectrum comparisons, including odd and even sizes. Maximum scaled direct-spectrum discrepancy was 3.366×10^{-15}; right and adjoint residuals were below 3.167×10^{-15}, original field residual below 1.074×10^{-16}. The chart/uniform-identity check had maximum scaled cancellation error 1.417×10^{-12}. The five-node reading is approximately

$$-2.9270509831, \quad \pm 4.9267960066\mathrm{i}, \quad -3.3541019662 \pm 10.0983504152\mathrm{i}.$$

These finite checks corroborate, but do not prove, all-n claims or certify intervals. Floating-point cell equality uses a tolerance; the two specified boundary obstructions are checked exactly. The saved JSON contains the actual counts and residuals; no historical developer statistics are claimed as independently reproduced.

Developer validation authentication. `output/p11/P11_validation_support.zip` has SHA-256

`3fc91d1788cbbbec0ecdd0f8df972481a71`
`33e6a87ce0f44bdcacf187d8ad18d.`

Its CRC and all 36 members were inventoried; the original remains unchanged. This establishes package identity only. The checker and historical counts were not rerun or adopted as proof. The audit's new script and results are independently authored. Compilation and visual inspection likewise do not supply mathematical verification.

10 Complete source-line review

The following ranges partition all 600 lines without gaps or overlaps. Every substantive range receives exactly one permitted disposition. Blank lines, labels and environment delimiters are attached to their neighboring substantive range. PASS means valid within the stated scope, not a proof of an excluded later result. All findings refer only to the authenticated hash.

Lines	Verdict	Finding
1–18	PASS	Version, inputs and exclusions match the manifest; no numerical premise.
19–31	PASS	Spectral scope is explicit; no crossing, orbit or propagation assertion.

Lines	Verdict	Finding
32–40	PASS	Positive equilibrium and effective coefficients use the canonical domain.
41–53	PASS	P04 phases, signed imaginary parts and unitary diagonalization agree exactly.
54–64	PASS	Complete representative maximizers include all ties and singleton classes.
65–79	PASS	Reachable bound proves a genuine pair; all even sizes exclude Nyquist.
80–96	PASS	Frequency factorization and both cosine-sign cases are exact.
97–103	PASS	Double-angle identity and positive sine justify every equivalence.
104–114	PASS	Strict means one complete representative, zero real part, nonzero frequency.
115–125	PASS	Exactly two distinct simple imaginary eigenvalues; every other label is left.
126–139	PASS	Finite positive gap, real dimensions and both inverse gates are correctly stated.
140–148	PASS	Representative-to-label transfer accounts for uniform and Nyquist modes.
149–159	PASS	Numerical simplicity follows from label exclusion and P04 multiplicity.
160–168	PASS	Noncritical set is nonempty; both critical-value comparisons are valid.
169–182	PASS	Same normalized Fourier column is the opposite-sign Hermitian adjoint.
183–191	PASS	Ordered real block and signed coefficient-plane angular velocity are correct.
192–198	PASS	Unitary diagonalization verifies the current pairing convention.
199–205	PASS	DFT orthogonality yields equal real/imaginary norms and a real basis.
206–214	PASS	Column signs are correct; negative signed frequency is handled explicitly.
215–223	PASS	Critical ties fail the single-pair hypothesis; real-part ties need not be eigenvalue ties.
224–234	PASS	Zero multiplicity is exactly the zero-label count; strict zero pair has real kernel dimension two.
235–244	PASS	P02 equilibrium chart is exactly the P05 open physical domain.
245–251	PASS	Diagonal simplification cancels the g terms; scalar cosine notation is distinguished.
252–261	PASS	R_j equals the selected modal real part with all physical coefficients.
262–272	PASS	Partial k derivative and solved graph have correct signs.
273–280	PASS	Regular surface retains k and a positivity, strict selection and nonzero frequency.
281–286	PASS	Rank-two assertion is conditional on existence and correct physical transfer.
287–301	PASS	P05 forces concavity; determinant is positive; transfer is local only.
302–305	PASS	Branch-coordinate derivative is expressly separated from the frozen physical path.
306–318	PASS	Frequency partials and negative nonzero two-by-two determinant are exact.
319–334	PASS	Cell criterion uses the correct lower midpoint and strict inequality.

Lines	Verdict	Finding
335–347	PASS	Fixed vertex/state iff claim is correct; both obstruction examples are valid.
348–361	PASS	Positive k forces w positive and z below B; all endpoints exist with correct indexing.
362–371	PASS	Both upper bounds exceed the lower bound; small positive x gives positive k.
372–380	PASS	Physical realization preserves effective coefficients and the equilibrium equation.
381–388	PASS	Large G enforces a positive and avoids at most one frequency zero; no universal existence claim survives.
389–399	PASS	P07 constants and frequency are conditional five-node inputs only.
400–412	PASS	Exact tuple and critical state are strictly positive and physically valid.
413–426	PASS	All five labeled eigenvalues, individual simplicity, ordering and exact gap are correct.
427–434	PASS	Second root lies strictly between zero and one.
435–442	PASS	Exact positivity and original-model residual are checked before spectral reduction.
443–452	PASS	Effective data and raw diagonal formula agree independently.
453–465	PASS	Radical real parts and strict noncritical difference are correct; table corroboration is redundant.
466–474	PASS	Frequency signs and exact positive bounds yield the complete complex spectrum.
475–484	PASS	P02 polynomial factors exactly; no P10 input is used.
485–492	PASS	Positive lower root and discriminant establish the upper-root identification.
493–504	PASS	Frozen g-path and unique local root near one are accurately scoped.
505–514	PASS	Explicit vector uses positive exponents, normalized pairing and correct adjoint sign.
515–526	PASS	Real block and both spectral-projector inverse sums are correct.
527–534	PASS	Mode-zero and mode-two denominators have exact values and are nonzero.
535–545	PASS	At X=1 the bracket derivative equals lambda_0; local positivity and root ordering persist.
546–555	PASS	Vector and denominator proofs introduce no tensors or nonlinear conventions.
556–563	PASS	H1-H2 are proved physical and original-field assertions.
564–565	PASS	H3 is local smooth upper-branch existence only.
566–569	PASS	H4-H6 assert the exact pair, individual simplicity and stable gap.
570–571	PASS	H7-H8 are exactly the two proved spectral exclusions.
572–573	PASS	H9 records established vectors, pairing and plane orientation.
574–575	PASS	H10 explicitly leaves transversality and path derivatives unestablished.
576–577	PASS	H11 explicitly leaves Lyapunov and nonlinear nondegeneracy unestablished.
578–580	PASS	H12 withholds periodic orbits, branch stability and rotating waves.
581–587	PASS	Ledger references correctly distinguish results from exclusions.

Lines	Verdict	Finding
588–592	PASS	Exact nonemptiness is analytical; computation is corroboration and C073 is unused.
593–596	PASS	B02/B06 spectral choices are scoped; B07 nonlinear normalization stays with P13.
597–600	PASS	First loss/open nonlinear regime remains unproved; proposed note is separate and frozen files untouched.

11 Request coverage and artifact inventory

The source-level ledger above is supplemented by this task-check mapping. The numbers refer to the supplied audit request, not P11 equation numbers.

Request items	Disposition and location
0–1	Current-version authentication and actual dependency use: Section 2 and the hash inventory.
2	All six layer verdicts are separately stated, with an overall verdict.
3–7	Orientation, reachability, frequency, strict definition and full label coverage: Section 3.
8–14	Simplicity, gap, real dimensions, inverse gates, adjoint, real block, ties and repeated zero: Section 4.
15–19	Equilibrium chart, real-part function, k graph, physical transfer and rank-two locus: Section 5.
20–29	Independent necessity/sufficiency, physical realization, frequency avoidance and exact obstructions: Section 6.
30–46	Original-field residual, exact spectrum, upper root, local branch, vectors, block and denominators: Section 7.
47–50	Literal H1–H12 ledger, separate clarification recommendation and P12/P13 boundaries: Section 8.
51–52	Analytical counterexample attempts and independent exact/finite checks: Section 9.
53	Complete 1–600 range partition: Section 10 and companion CSV.
54–58	Required verdicts, exact slug, audit-only ZIP, source/PDF, preservation and explicit closure handoff.

The audit-only archive is named exactly `cyclic_network_project_A11_audit_only.zip`, with sole project root `cyclic_network_project/`. It contains this standalone audit source, its PDF reading copy and `audits/A11_support/` evidence only. It contains no replacement P11 source and is not a full-project snapshot. The support files record the audited version; all requested input hashes; all 200 original project hashes; archive identity and CRC checks; independent checker and results; the 646-cell realizability table; complete line review; clarification review; validation provenance; and preservation/build records. The original draft archives and every pre-existing project file are retained.

12 Authenticated input hashes

The full machine-readable inventory includes byte counts, line counts and proof-body hashes. The following human-readable list records all 31 inputs. Each displayed SHA-256 is split only for page width; concatenate its halves.

project_manifest.md
e57b131d77a821547cc30a2b1efb21e1
4a71548d6d40668a50140d9bc8e17272
claim_registry.md
4e1a7fd2a38b4364e98570c55b211c27
7348082693ab1e56367222e4f4d37b8f
frozen_theorem_package.md
724adf3d04a7f9661a2e53fbae359192
e8a5a5ee56737bdca2379209b556ea63
final_novelty_verdict.md
5fa8ddd5ada9628b104dba68dd535356
442c5e62ec45b72f3c2c354dec6037ea
novelty_decisions.md
8fd079fb108b93c4504bc2b527de11c1
f7e867a967c7cef0d90e1ff7a0a9175e
open_decisions.md
506c634c9958d13b4d6ef4080ebe214a
f5fd0c77d5dc8fe6f1b91d2750ffab59
02_theorem_dependency_map.tex
a1a4435f36fc1d1b6f0b2c5226a91328
3c76658d55b96493e6c50a8693831cac
proofs_compendium.tex
88afbd023e84e6396f34b785fe54dffe
4fcc398796e6ff3ec6da45300776b62f
shared_macros.tex
f4a18a440e1214b619bf18f4a81fee50
7239460ca796d8ca3916791069d6a7ad
00_canonical_model.tex
c209981cb2924e8a782f24aa2904e8c8
5a105ff99319dd6aada12f48a351c2a4
01_notation_and_assumptions.tex
97d993c6d3986524dbcc29659ee73d25
7be1b03a35c44401e8ef4534303eeb91
proofs/P01_wellposedness_and_absorbing_box.tex
0d6950821aa83f3d09e8bfe84dab2b0b
5020aa6e7881cf6928e984bff1890f1e
audits/A01_wellposedness_and_absorbing_box.tex
8d5c1b2ec782dbe49b07d2bbe529ee47
4d4c142695c53b523ccb41d63bb33098
proofs/P02_synchronous_subspace_and_equilibria.tex
8e057c2cd009659eee45906844e25f4f
38c156e5eb2e4130e824aa5e8299c51a

audits/A02_synchronous_subspace_and_equilibria.tex
ca3d723d9fcfac14f5129583441e611a
65e41848ec44f3b5e3a0072f795d84ac
proofs/P03_origin_and_linearization.tex
6c3c9bfb2307bdc043eb7f6a874cb222
e2ac2ce0750b339181f0088dd0bd859e
audits/A03_origin_and_linearization.tex
e1008f98ba9667b92b8d74f1bc7d21e0
21414b454ac976ae0db1500b4b7ef8d1
proofs/P04_arbitrary_nq_fourier_spectrum.tex
1331c8598acd0481046250d918cac028
ca6d13d2604612448c3459e864321c58
audits/A04_arbitrary_nq_fourier_spectrum.tex
8a57079241269c92a3c3e8dc4dccf859
2da2d1a636fca0c0990a4f318396d869
proofs/P05_q2_mode_selection.tex
054a047a55ef37815485f6c225be2f63
22ca8cf1c272614dd1dc4bb10ae8e1dd
audits/A05_q2_mode_selection.tex
0f7e9836b91009a201185a4acbbb9971
5f3c65df715e405962b7d6852ff27031
proofs/P07_five_node_specialization.tex
1258641da2b1b729523da6262cf75345
f5ef1eabf7a9f693a8b13dd34deec393
audits/A07_five_node_specialization.tex
b3923d6520cc249c508b80dd06a2b515
f054e1510e255224a864e986cd38ce4b
proofs/P08_matched_bilinear_analysis.tex
81d273f0bad1714d2470e9fc826f0819
e9dde367a9c42633705a275b404e5454
audits/A08_matched_bilinear_analysis.tex
87cd4a03061648eb4cf1c65d1afa47c7
132c1f7eed10d546784ee166c8e5c935
proofs/P09_transcritical_bifurcation.tex
5d357a5be97dfbad62a0107ca2e010ff
97e0c11a719ee850df69411275f11ff8
audits/A09_transcritical_bifurcation.tex
1234e02d7cf620511bfd7b9306dbb72a
c68929ef01e8c3ba70bd3615bcd2b943
proofs/P10_saddle_node_and_bistability.tex
37542cc5409d7db33ca183869933fbdb
ad673628f69f6988f2778a84bc84094f
audits/A10_saddle_node_and_bistability.tex
ff2ebb0f84ae79fc9aad012a3945ef23
6e8ccdac9f47d64fd708e4dc66963b19
proofs/P11_hopf_spectral_conditions.tex
4be8e9b1e64c022b635f3cca9f7f49ff
bc5358a2250dda1715cb60e82d9b073b

`PROPOSED_P11_scope_clarification.md`
`027c130a0ed4e472335746689f63a194`
`d286fd01648558b04feb18d1f3132b0a`

13 Closure handoff

OVERALL VERDICT: VERIFIED. Required corrections: none. There are no E01, E02 or other required mathematical or wording issues to close. This verdict applies only to the exact P11 v1.0 hash identified at the start of this audit. The mathematical body requires no change.

The **ORIGINAL P11 development chat may perform administrative closure without changing the mathematical body**. It should preserve this exact version identity, record the independent A11 verdicts in the appropriate lifecycle records, retain the scope exclusions and validation-provenance distinctions, and handle the separately recommended planning clarification explicitly. This audit does not itself mark P11 VERIFIED and does not edit P11, its manifest, the registry, the compendium, or a frozen file.

After successful correction/closure, the original P11 development chat must create `cyclic_network_project_P11_audited.zip` as the **COMPLETE full-project snapshot**. That archive is not created in this audit chat. The existing `cyclic_network_project_P11_draft.zip` must remain unchanged. The deliverable here is solely the audit-only archive.

P12 is cleared to begin after administrative closure. This is permission to undertake its scoped proof work, not certification of a P12 result. P12 must analyze the actual frozen physical path and account for equilibrium motion; this audit computes no derivative along it. P13 must still establish nonlinear nondegeneracy with reconciled conventions before any assembled Hopf, periodic-orbit or branch-stability conclusion. First-loss and rotating-wave claims remain subject to their separate outstanding gates. Neither P12 nor P13 is begun here.